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RESUME

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Personal Information

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G-2, Prestige Marrakesh Apartments, 6/1, St Marks Road, Bengaluru, Karnataka-560001, INDIA

Date of birth: September 21, 1948, born in Puttur, Dakshina Kannada District, Karnataka, India

Citizenship: Indian, Passport number: Z 5017336 valid up to 26-6-2028

Wife: Prathibha Karanth, Ph.D (Speech and Language Sciences)

Daughter: Krithi K. Karanth, Ph.D (Conservation Biology)

Language skills: Read, write and speak: English, Kannada (mother tongue), Hindi and Tulu.

Comprehension: Tamil, Telugu, Kodava, Konkani

Professional Positions held

Emeritus Director, Centre for Wildlife Studies (1984-to date)

Director, India Program, Wildlife Conservation Society (1988-2010)

Director for Science-Asia, and, Director, India Program, Wildlife Conservation Society (2012-2017)

Educational Background

1993: Ph. D degree in Applied Zoology, Mangalore University, India. Thesis title: "*Predator-prey relationships among the large mammals of Nagarhole National Park (India)*".

1988: M. S. degree in Forest Resources and Conservation, University of Florida, Gainesville, USA. Thesis title: "*Population structure, density and biomass of large herbivores in a South Indian tropical forest*".

1971: Bachelor's degree in Mechanical Engineering, Karnataka Regional Engineering College (Now National Institute of Technology), Surathkal, Mangaluru, India.

Special Skills and Interests

Biology and management of large carnivores such as tigers and leopards including knowledge of

methods of their capture, chemical immobilization and radio-tracking

Wildlife Population sampling, estimation, modeling and analysis with special expertise in camera- trap surveys, line transect sampling and occupancy modeling approaches

Conservation policy, science-based conservation advocacy and science popularization

Professional Affiliations

Academic and Scientific Positions Held

Fellow, Indian Academy of Sciences, Bangalore, India
Adjunct Professor, University of Florida, Gainesville, USA
Adjunct Professor, University of Minnesota, Minneapolis, USA
Adjunct Professor, Tata Institute of Fundamental Research (NCBS), Bengaluru (2004-2020)
Adjunct Professor, Kasetsart University, Bangkok, Thailand (2023-present)
Recognized Doctoral Guide, Manipal Academy of Higher Education (MAHE), Manipal, India
Associate, National Institute of Advanced Studies (NIAS), Bangalore, India
Life Member, Bombay Natural History Society
Associate Editorial Board, *PLoS One* (2013-2016)
Associate Editor, *Journal of Applied Ecology* (2005-2008)
Member, Editorial Board, *Oryx* (2004-present)
Member, Editorial Board, *Animal Conservation* (1996-2001)
Member, Editorial Board, *Conservation & Society* (2004-2007)
Member, Editorial Board, *Journal of Bombay Natural History Society* (2005-2019)

Positions Held on Governing Boards of Organizations and Institutions

Member, Steering Committee, Project Tiger, Government of India (1995-2005)
Member, Forest Advisory Committee, Government of India (2008-2012).
Member, National Tiger Conservation Authority, Government of India (2009-2015)
Member, Governing Council, Wildlife Institute of India (2009-2015)
Member, Karnataka State Wildlife Advisory Board (1983-2013, intermittently)
Member, Advisory Board, Liz Claiborne and Art Ortenberg Foundation, USA (2013-2024)
Member, Science Council, Panthera Foundation, USA (2007-2012 and 2019-present)
Member, Board of Trustees, Worldwide Fund for Nature-India (WWF), (2003-2006)

Awards, Honors and Recognitions

2021: Distinguished Alumnus Award, The University of Florida, Gainesville, USA

2018: The George Schaller Award, Wildlife Conservation Society, USA

2018: Distinguished Alumnus Award, St. Aloysius College, Mangalore, India.

2018: Thrive Award, Seattle Zoological Society (Jointly with Krithi Karanth)

2018: Short listed for the ACE award of Banovich Foundation

2013: Short listed for the Indianapolis Award

2012: Padma Shri, National Civilian Honor Conferred by the President of India

2011: Rajyprashasthi State Award conferred by the Government of Karnataka.

2010: Distinguished Alumnus, National Institute of Technology, Surathkal

2009: *Salim Ali National Award* for lifetime achievement, Bombay Natural History Society
2008: **Elected Fellow of the Indian Academy of Sciences**
2007: **J. Paul Getty Award for Global Conservation Leadership, conferred by WWF -USA**
2006: **Earthcare International award conferred by the Sierra Club, USA**

Major Scientific Talks and Presentations

1990-2022: Made invited presentations at prestigious academic and research institutions including Oxford University, Cambridge University, Princeton University, Columbia University, Duke University, University of Florida, University of Minnesota, University of Massachusetts, California Academy of Sciences, Royal Geographic Society, Smithsonian Institution, US Fish & Wildlife Service, US Geological Survey, Explorers Club, Zoological Society of London, National Centre for Biological Sciences-TIFR, Jawaharlal Nehru Centre for Advanced Studies, Indian Institute of Science, Indian Statistical Institute, Manipal Academy of Higher Education.

Presentations at professional meetings, including at The Wildlife Society, Society for Conservation Biology, International Congress of Ecology and International Congress of the Society of Mammalogy and the International Wildlife Management Congress.

Coverage of Professional Work in the Media

1990-2020: My Scientific and conservation work has received wide coverage in Indian and international media including several television documentaries broadcast on *BBC, CNN, PBS, CBC, National Geographic* and *Discovery* Television Channels as well as national and global print media including *New York Times, Guardian, Time, Nature, Science, Scientific American, New Scientist, BBC Wildlife* and *BBC Futures*.

I have published over 100 popular articles, blogs and interviews on wildlife science and conservation issues in English and Kannada languages.

Scientific Research Projects Conducted (1986 to 2018)

Co-Principal investigator, (with Dr. Melvin Sunquist), CWS Project titled "*Ecological relations and resource use in the carnivore-herbivore community of Nagarahole National Park*" under funding from US Fish and Wildlife Service and Government of Karnataka, 1986-1989.

Co-Principal investigator, (with Dr. Melvin Sunquist), CWS Project titled "*Ecology and management of large carnivores*" in Nagarahole National Park, with funding from US Fish and Wildlife Service, The Wildlife Conservation Society, New York, and National Geographic Society, 1989-1993.

Principal investigator, CWS Project titled "*Ecological status and conservation of tigers in India*" supported by US-Fish and Wildlife Service, US-India Fund and Wildlife Conservation Society, New York, 1995-1999.

Principal Investigator (with The Chief Wildlife Warden, Karnataka), CWS Project titled "*Karnataka Tiger Conservation Project*" supported by Wildlife Conservation Society, National Fish and Wildlife Foundation-Save the Tiger Fund and Exxon Foundation, USA, 1998-2000.

Principal investigator, CWS Project titled "*Distribution and dynamics of tiger and prey populations in Maharashtra*" supported by US-Fish and Wildlife Service and Wildlife Conservation Society, New York, 2000-2004.

Principal investigator, CWS Project titled "*Distribution and dynamics of tiger and prey populations in Karnataka*" by US-Fish and Wildlife Service and Wildlife Conservation Society, New York, 2003-2007.

Principal investigator, CWS Project titled "*Metapopulation dynamics of tigers in the Malenad-Mysore Landscape of Karnataka*" supported by Department of Science & Technology and Department of Biotechnology, Government of India, Wildlife Conservation Society, New York, and US Fish and Wildlife Service, 2007-2019.

SUPERVISION OF POST-GRADUATE STUDENTS

Supervised eight Master's Degree students at the Post-Graduate Program in Wildlife Biology and Conservation, at National Centre for Biological Sciences, Bangalore, Manipal University, India and one Master's student Jai Ranganathan at the University of Minnesota. USA.

Supervised four Ph.D. students (GV Reddy, N. Samba Kumar, Vidya Athreya and Devcharan Jathanna at Manipal University, India

Co-supervised: PhD student Arjun Gopaldaswamy (Oxford, UK), PhD Chandan Pandey (Manipal University, India).

BOOKS PUBLISHED

Popular Books in English

2022. *Among Tigers: Fighting to Bring Back Asia's Big Cats*. Chicago Review Press, USA

2021. *Growing up Karanth* (co-authored with Malavika Kapur and Kshama Rao). West Land Publications, India

2006. *A View from the Machan*, Permanent Black, India. **Also translated into Chinese Mandarin.**

2006. *Tiger Tales*, (Edited Anthology), Penguin Press, India.

2001. *The Way of the Tiger*, Voyageur Press, USA and Colin Baxter, UK & Universities Press India. **This book has been translated into Chinese Mandarin, Telugu, Kannada and Tamil.**

Popular Books in Kannada Language

2016. *Karanthara Suliyalli* (Kannada translation of *Growing up Karanth* by Vishweshwar Bhat)

2007. *Hulirayana Akashavaani* (*A Tiger's Broadcast*, with TS Gopal)

2005. Huliya Baduku (*Life of the Tiger*)

2000. Kaadu Pranigala Jadinalli (*Tracking Wild Animals*)

1983. Aranya Matthu Samaja (*Forests and Society*)

Popular Articles in English

Popular Articles in Kannada

Scientific Books and Monographs

2021. *Spatial Dynamics and Ecology of Large Ungulate Populations in Tropical Forests of India* (Co-authored with N Samba Kumar, James D Nichols, Srinivas Vaidyanathan, Beth Gardner and Jagdish Krishnaswamy), Springer-Nature, Singapore.

2017. *Methods for Monitoring Tiger and Prey Populations*, (Co-edited with James D Nichols) Springer-Nature, Singapore.

2017. *Science and Conservation of Animal Populations*. Natraj Publishers, New Delhi, India

2016. *Recovering Biodiversity in Indian Forests* (Co-authored with G Vishwanatha Reddy, N Samba Kumar, Jagdish Krishnaswamy and Krithi K Karanth), Springer Nature, Singapore.

2011. *Camera traps in Animal Ecology* (Co-edited with Allan F. O'Connell and James D Nichols), Springer Nature, Tokyo, Japan.

2010. *Science of Saving Tigers*, Orient Blackswan, Hyderabad, India.

2002. *Monitoring Tigers and their Prey* (Co-edited with James Nichols), Centre for Wildlife Studies, Bangalore.

Scientific Articles and Papers

Karanth K.U., Kumar N.S., **Karanth, K.K.** (2020). Tigers Against the Odds: Applying Macro-Ecology to Species Recovery in India. *Biological Conservation*: <https://www.sciencedirect.com/science/article/pii/S0006320720309046?via%3Dihub>

Jayadevan, A., Nayak, R., **Karanth, K. K.**, Krishnaswamy, J., DeFries, R., **Karanth, K.U.**, Vaidyanathan, S. (2020). Navigating paved paradise: Evaluating landscape permeability to movement for large mammals in two conservation priority landscapes in India. *Biological Conservation*, 247,108613. <https://doi.org/10.1016/j.biocon.2020.108613>

Jathanna, D., Kumar, S., N., **Karanth, K.**, U., (2020) Summary and highlights of small carnivore photo-captures during a field season in the central Western Ghats, India. *Small Carnivore*

Conservation, vol. 58, e58017. ISSN 1019-5041.

Kumar, A.V., **Karanth, K.U.**, Jathanna, D. (2020) Tigers and leopards coexist despite similarities in space use and habitat selection. *Cat News* (Spring).

Nayak, R., **Karanth, K. K.**, Dutta, T., De Fries, R., **Karanth, K.U.**, Vaidyanathan, S. (2020). Bits and pieces: Forest Fragmentation by Linear Intrusions in India. *Land Use Policy*. doi:<https://doi.org/10.1016/j.landusepol.2020.104619>

Athreya, V., Isvaran, K., Odden, M., Linnell, J. D., Kshetry, A., Krishnaswamy, J., **Karanth, K.U.** (2020). The impact of leopards (*Panthera pardus*) on livestock losses and human injuries in a human-use landscape in Maharashtra, India. *PeerJ*. doi: <https://peerj.com/articles/8405/>

Gopalaswamy M. A., Karanth K. U., Delampady M., Stenseth C. N. 2019. How sampling- based overdispersion reveals India's tiger monitoring orthodoxy. *Conservation Science and Practice*. doi: <https://conbio.onlinelibrary.wiley.com/doi/full/10.1111/csp2.128>

Dey, S., Delampady, M., **Karanth, K.U.** and Gopalaswamy, A.M. (2019). A spatially explicit capture-recapture model for partially identified individuals when trap detection rate is less than one. *Calcutta Statistical Association (Bulletin)*, Vol. 71 (1):1-20.

Ahrestani, F., Kumar, N.S., Vaidyanathan, S., Hiby, L., Jathanna, D., **Karanth, K.U.** (2018) Estimating densities of large herbivores in tropical forests: Rigorous evaluation of a dung-based method. *Ecology and Evolution*. doi: <https://doi.org/10.1002/Ece3.4227>

Gardner, B., Sollman, R., Kumar, N.S., Jathanna, D., **Karanth, K.U.** (2018) State space and movement specification in open population spatial capture–recapture models. *Ecology and Evolution*. doi: <https://doi.org/10.1002/Ece3.4509>

Srivathsa, A., Puri, M., Kumar, N.S., Jathanna, D., **Karanth, K.U.** (2017) Substituting space for time: Empirical evaluation of spatial replication as a surrogate for temporal replication in occupancy modelling. *Journal of Applied Ecology*. 55: 754–765. doi: <https://doi.org/10.1111/1365-2664.13005>

Karanth, K.U., Nichols, J.D., Goodrich, J.M., Reddy, G.V., Mathur, V.B., Wibisono, H.T., Sunarto, S., Pattanavibool, A., & Gumal, M. (2017) Role of monitoring in global tiger conservation. *Methods for Monitoring Tiger and Prey Populations* (eds K.U. Karanth & J.D. Nichols), pp. 1-14. Springer Nature, Singapore.

Karanth, K.U., Goodrich, J.M., Miquelle, D.G., Ramesh, K., Harihar, A., Pattanavibool, A., Sunarto, S., Rayan, M.D. & Thapa, K. (2017) Tiger ecology in relation to monitoring issues. *Methods for Monitoring Tiger and Prey Populations* (eds K.U. Karanth & J.D. Nichols), pp.15-34. Springer Nature, Singapore.

Nichols, J.D., **Karanth, K.U.**, Gopalaswamy, A.M. & Delampady, M. (2017) Animal population monitoring: A unified conceptual framework. *Methods for Monitoring Tiger and Prey Populations* (eds K.U. Karanth & J.D. Nichols), pp.35-46. Springer Nature, Singapore.

Nichols, J.D., Goswami, V.R., Parameshwaran, R., Dey, S. & **Karanth, K.U.** (2017) Concepts:

assessing tiger habitat occupancy dynamics. *Methods for Monitoring Tiger and Prey Populations* (eds K.U. Karanth & J.D. Nichols), pp.47-70. Springer Nature, Singapore

Kumar, N.S., Puri, M., Miquelle, D.G, Pattanavibool, A., Harihar, A., Sunarto, S., Vaidyanathan, S. & **Karanth, K.U.** (2017) Field practices: Assessing tiger habitat occupancy dynamics. *Methods for Monitoring Tiger and Prey Populations* (eds K.U. Karanth & J.D. Nichols), pp.71-88. Springer Nature, Singapore.

Strindberg, S., Kumar, N.S., Thomas, L. & Goswami, V.R. (2017) Concepts: Estimating abundance of prey using line transects sampling. *Methods for Monitoring Tiger and Prey Populations* (eds K.U. Karanth & J.D. Nichols), pp.89-120. Springer Nature, Singapore.

Royle, J.A., Gopalaswamy, A.M., Dorazio, R.M., Nichols, J.D., Jathanna, D., Parameshwaran, R. & **Karanth, K.U.** (2017) Concepts: assessing tiger population dynamics using capture-recapture sampling. *Methods for Monitoring Tiger and Prey Populations* (eds K.U. Karanth & J.D. Nichols), pp.163-190. Springer Nature, Singapore.

Karanth, K.U., Nichols, J.D., Harihar, A., Miquelle, D.D., Kumar, N.S. & Dorazio, R.M. (2017) Field practices: Assessing tiger population dynamics using photographic captures. *Methods for Monitoring Tiger and Prey Populations* (eds K.U. Karanth & J.D. Nichols), pp.191-224. Springer Nature, Singapore.

Dorazio RM, **Karanth, K.U.** (2017) A hierarchical model for estimating the spatial distribution and abundance of animals detected by continuous-time recorders. *PLoS ONE* 12(5): e0176966.

Dey, S., Delampady, M., Ravishankar, P., Kumar, S.N., Srivathsa, A., and **Karanth, K.U.** (2017). Bayesian Methods for Estimating Animal Abundance at Large Spatial Scales Using Data from Multiple Sources. *Journal of Agricultural Biological and Environmental Statistics*. doi: 10.1007/s13253-017-0276-7

Karanth, K.U., Srivathsa, A., Vasudev, D., Puri, M., Parameshwaran, R., and Kumar, S.N. (2017). Spatio-temporal interactions facilitate large carnivore sympatry across a resource gradient. *Proceedings of the Royal Society-B*, 284: 20161860

Srivathsa, A., Kumar, S.N. and **Karanth, K.U.** (2017). Home range size of the dhole from camera-trap surveys. *Canid Biology & Conservation*, 20(1), 1-4

Muthanna, P.M., Vasudev, D. & **Karanth, K.U.** (2017) Leveraging public funding to secure a protected area in the Western Ghats, India. *Oryx*, 51(1), 16-17.

Karanth, K.U. (2016). Tracking Tigers: The trouble with Tiger Numbers (2016). *Scientific American*. July, 315 (1).

Duangchantrasiri, S., Umponjan, M., Simcharoen, S., Pattanavibool, A., Chaiwattana, S., Maneerat, S., Kumar, N. S., Jathanna, D., Srivathsa, A., **Karanth, K.U.** (2016). Dynamics of a low-density source population of tigers in Southeast Asia in the context of improved law enforcement. *Conservation Biology*. DOI: 10.1111/cobi.12655

Goodrich, J., Lynam, A., Miquelle, D., Wibisono, H., Kawanishi, K., Pattanavibool, A., Htun, S.,

Tempa, T., Karki, J., Jhala, Y. & **Karanth, U.** (2015). *Panthera tigris*. The IUCN Red List of Threatened Species 2015: e.T15955A50659951

Creel, S., Becker, M., Christianson, D., Dröge, E., Hammerschlag, N., Hayward, M. W., **Karanth, K.U.**, Loveridge, A., Macdonald, D. W., Matandiko, W., M'soka, J., Murray, D., Rosenblatt, E., Schuette, P. (2015). Questionable Policy for Large Carnivore Hunting. *Science*. DOI: 10.1126/science.aac4768

Hayward, M. W., Boitani, L., Burrows, N. D., Funston, P. J., **Karanth, K.U.**, MacKenzie, D. I., Yarnell, R. W. (2015). Ecologists need robust survey designs, sampling and analytical methods. *Journal of Applied Ecology*, 52 (2): 286-290. DOI: 10.1111/1365- 2664.12408

Puri, M., Srivathsa, A., **Karanth, K. K.**, Kumar, N. S., **Karanth, K.U.** (2015). Multiscale distribution models for conserving widespread species: The case of sloth bear *Melursus ursinus* in India. *Diversity and Distributions*, 1–14. DOI: 10.1111/ddi.12335

Gopalaswamy, A. M., Delampady, M., **Karanth, K.U.**, Kumar, N. S., Macdonald, D. W. (2015). An examination of index-calibration experiments: Counting tigers at macroecological scales. *Methods in Ecology and Evolution*, 6 (9): 1055-1066. DOI: 10.1111/2041-210X.12351

Jathanna, D., **Karanth, K.U.**, Kumar, N. S., Goswami, V. R., Vasudev, D., **Karanth, K. K.** (2015). Reliable monitoring of elephant populations in the forests of India: Analytical and practical considerations. *Biological Conservation*, 187, 212-220. Doi:10.1016/j.biocon.2015.04.030

Athreya, V., Srivathsa, A., Puri, M., **Karanth, K. K.**, Kumar, N. S., **Karanth, K.U.** (2015). Spotted in the News: Using Media Reports to Examine Leopard Distribution, Depredation, and Management Practices outside Protected Areas in Southern India. *PLoS ONE*, 10(11), e0142647. DOI: 10.1371/journal.pone.0142647

Lakshminarayanan, N., **Karanth, K. K.**, Goswami, V. R., Vaidyanathan, S., **Karanth, K.U.** (2015). Determinants of dry season habitat used by Asian elephants in the Western Ghats of India. *Journal of Zoology*. DOI: 10.1111/jzo.12298

Jathanna, D., **Karanth, K.U.**, Kumar, N. S., **Karanth, K. K.**, Goswami, V. R. (2015): Patterns and determinants of habitat occupancy by the Asian elephants in the Western Ghats of Karnataka, India. *PLoS ONE*, 10(7): e0133233. DOI: 10.1371/journal.pone.0133233.

Srivathsa, A., Parameshwaran, R., Sharma, S., **Karanth, K.U.** (2015). Estimating population sizes of leopard cats in the Western Ghats using camera surveys. *Journal of Mammalogy*. DOI: <http://dx.doi.org/10.1093/jmammal/gyv079>

Athreya, V., Odden, M., Linnell, J. D. C., Krishnaswamy, J., **Karanth, K.U.** (2014). A cat among the dogs: Leopard *Panthera pardus* diet in a human-dominated landscape in western Maharashtra, India. *Oryx*, 1 - 7. DOI: <http://dx.doi.org/10.1017/S0030605314000106>

Mondol, S., Kumar, N. S., Gopalaswamy, A., Sunagar, K., **Karanth, K.U.**, Ramakrishnan, U. (2014). Identifying species, sex and individual tigers and leopards in the Malenad- Mysore Tiger Landscape, Western Ghats, India. *Conservation Genetics Resources*, 1-9. DOI:

Karanth, K.U., Kumar, N., Vasudev, D. (2014). Photographic database informs management of conflict tigers. *Oryx*, 48(04): 484-484. DOI: <http://dx.doi.org/10.1017/S0030605314000532>

Ahrestani, F. and **Karanth, K.U.** (2014) Gaur *Bos gaurus*. In Ecology, Evolution and Behavior of Wild Cattle: Implications for Conservation. M. Melletti and J. Burton (Editors). Cambridge University Press, UK.

Srivathsa, A., **Karanth, K.K.**, Jathanna, D., Kumar, N.S., **Karanth, K.U.** (2014). On a Dhole Trail: Examining Ecological and Anthropogenic Correlates of Dhole Habitat Occupancy in the Western Ghats of India. *PLoS ONE* 9(6): e98803. DOI: 10.1371/journal.pone.0098803

Athreya, V., Navya, R., Punjabi, G. A., Linnell, J. D. C., Odden, M., Khetarpal, S., **Karanth, K.U.** (2014). Movement and activity pattern of a collared tigress in a human- dominated landscape in central India. *Tropical Conservation Science* Vol.7 (1): 75-86.

Jones, J. P. G., Asner G. P., Butchart S. H. M., **Karanth, K.U.** (2013). The ‘why’, ‘what’ and ‘how’ of monitoring for conservation. In Key Topics of Conservation Biology 2. Oxford University Press (Editors: David MacDonald and Katherine Wills). Pages 329-343.

Athreya, V., Odden, M., Linnell, J. D. C., Krishnaswamy, J., **Karanth, K.U.** (2013). Big cats in our backyards: Persistence of large carnivores in a human dominated landscape in India. *PLoS ONE*: 8(3): e57872. doi: 10.1371/journal.pone.0057872

Karanth, K.U. (2013). Tiger. In Mammals of South Asia. Volume 1. University Press (Editors: AJT Johnsingh and Nima Manjrekar). Pages 596-614.

Karanth, K.U. (2013). Leopard. In Mammals of South Asia. Volume 1. University Press (Editors: AJT Johnsingh and Nima Manjrekar). Pages 559-574.

Karanth, K.U., Gopalaswamy, A. M., **Karanth, K. K.**, Goodrich, J., Seidensticker, J., Robinson, J. G. 2013. Sinks as saviors: Why flawed inference cannot assist tiger recovery. *Proceedings of the National Academy of Science, USA*. doi: 10.1073/pnas.1216623110.

Karanth, K.U., Kumar, N. S., Nishant, M. S. 2012. Estimating distribution and abundances of elephant populations from sign surveys at landscape scale using occupancy modeling: field methods. In Monitoring Elephant Populations and Assessing Threats: A Manual for Researchers, Managers and Conservationists. University Press (Editor: S. Hedges). Pages 249-293.

Karanth, K.U., Hedges, S., Kumar, N. S., Goswami, V. R., Nichols, J. D. 2012. Estimating abundance and other demographic parameters in elephant populations using capture– recapture sampling: field practices. In Monitoring Elephant Populations and Assessing Threats: A Manual for Researchers, Managers and Conservationists. University Press (Editor: S. Hedges). Pages 214-238.

Hedges, S., Kumar, N. S., Nishant, M. S., **Karanth, K.U.** 2012. Estimating density and abundance of elephants from sightings along line transects: field methods. In Monitoring Elephant Populations and Assessing Threats: A Manual for Researchers, Managers and

Conservationists. University Press (Editor: S. Hedges). Pages 151-162.

Karanth, K.U., Hedges, S., Kumar, N. S., Nichols, J. D. 2012. Estimating distribution and abundances of elephant populations from sign surveys at landscape scale using occupancy modeling: statistical concepts. In *Monitoring Elephant Populations and Assessing Threats: A Manual for Researchers, Managers and Conservationists*. University Press (Editor: S. Hedges). Pages 136-150.

Karanth, K.U., Nichols, J. D., Hedges, S. 2012. Estimating abundance and other demographic parameters in elephant populations using capture-recapture sampling: statistical concepts. In *Monitoring Elephant Populations and Assessing Threats: A Manual for Researchers, Managers and Conservationists*. University Press (Editor: S. Hedges). Pages 112-136.

Karanth, K.U., Nichols, J. D. 2012. Wildlife Population Monitoring: A Conceptual Framework. In *Monitoring Elephant Populations and Assessing Threats: A Manual for Researchers, Managers and Conservationists*. University Press (Editor: S. Hedges). Pages 1-8.

Gopalaswamy, A. M., Royle, J. A., Hines, J. E., Singh, P., Jathanna, D., Kumar, N. S., **Karanth, K.U.** (2012). Program SPACECAP: software for estimating animal density using spatially explicit capture-recapture models. *Methods in Ecology and Evolution*, 3 (6): 1067- 1072. DOI: 10.1111/j.2041-210X.2012.00241.x

Karanth, K.U., **Karanth, K. K.** 2012. A tiger in the drawing room: Can luxury tourism benefit wildlife. *Economic and Political Weekly* Sept 22nd. 38-43.

Gopalaswamy, A. M., **Karanth, K.U.**, Kumar, N. S., Macdonald, D. W. 2012. Estimating Tropical Forest Ungulate Densities from Sign Surveys using Abundance Models of Occupancy. *Animal Conservation* doi:10.1111/j.1469-1795.2012.00565.x

Gopalaswamy, A. M., Royle, J. A., Delampady, M., Nichols, J. D., **Karanth, K.U.**, Macdonald, D. W. 2012. Density Estimation in Tiger Populations: Combining Information for Strong Inference. *Ecology* 93:1741-1751.

Goswami, V.R., Laretta, M., Madhusudan, M.D., **Karanth, K.U.** 2012. Optimizing individual identification methods and sampling effort for estimation of Asian elephant abundance using photographic capture-recapture sampling. *Animal Conservation*. doi:10.1111/j.1469-1795.2011.00501.x

Pollock, K.N., Nichols J. D., **Karanth K.U.** 2012. Estimation of Demographic Parameters. In *Carnivore Ecology and Conservation: A handbook of Techniques*. Oxford University Press (Editors: L. Boitani and R. A. Powell). Pages: 169-193.

Karanth, K.U., Gopalaswamy, A. M., Kumar, N. S., Delampady, M., Nichols, J. D., Seidensticker, J., Noon, B. R., Pimm, S. L. 2011. Counting India's Wild Tigers Reliably. *Science* 332: 791.

Karanth, K.U., Gopalaswamy, A. M., Kumar, N. S., Vaidyanathan, S., Nichols, J. D., MacKenzie, D. I. 2011. Monitoring Carnivore Populations At The Landscape Scale. *Journal of*

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